Lecture II
Moran's I and Geary's C

STAT 679: Statistical Methods for Spatial Lattice Data-Spring 2026
University of Wisconsin-Madison

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Outline

1. Spatial Correlation
   - Review of Correlation
   - Moran's I
   - Geary's C
   - Spatial Correlation in the Context of Linear Regression: Examples
2. R-Demo
3. Assignments
   - Assignment 1 Solution
   - Assignment 2

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Recall..

- Spatial data, Lattice data, and spatial patterns in data
- Different types of spatial neighbors and join tests (BB, BW, WW)

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Brief Review of Population Correlation

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- Population correlation coefficient between two random variables X and Y is defined as

$$\rho = \mathrm{Cor}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}.$$

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  Where:

  - ρ is a measure of **linear relationship** between X and Y.
  - $-1 \leq \rho \leq 1$.
  - $\rho = 1$ indicates perfect positive correlation.
  - $0 < \rho < 1$ indicates some positive correlation.
  - $\rho = 0$ indicates no linear relationship.
  - $-1 < \rho < 0$ indicates some negative correlation.
  - $\rho = -1$ indicates perfect negative correlation.

Brief Review of Population Correlation (cont'd)

Brief Review of Population Correlation (cont'd)

Relations among Var, Cov, and Cor are

1. $$\mathrm{Var}(X) = E[(X - \mu_X)^2] = \sigma_X^2.$$
2. $$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)].$$
3. $$\mathrm{Cov}(X, X) = E[(X - \mu_X)^2] = \mathrm{Var}(X).$$
4. $$\mathrm{Cor}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$
5. **If X and Y are independent**, then $\mathrm{Cov}(X, Y) = 0$ and $\mathrm{Cor}(X, Y) = 0$.

Brief Review of Sample Correlation

Brief Review of Sample Correlation

- Based on data $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the sample correlation coefficient is defined as

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})/(n-1)}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2/(n-1)}\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2/(n-1)}}$$

$$= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

  and estimates the population correlation coefficient ρ.

- Note the symmetry between x and y in r.
- Remark:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = r \times \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$

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Moran's I

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$$I = \frac{n}{s_0} \times \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

where

$$s_0 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}.$$

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- Moran's I is a special case of the general form

$$G = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} y_{ij}$$

  with $y_{ij} = (x_i - \bar{x})(x_j - \bar{x})$ and some rescaling.
- Moran's I looks like a correlation and is a measure of similarity.

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- Moran's I looks like a correlation and is a measure of similarity.
- Under CSR, $E(I) = -1/(n-1)$ and the variance has a complicated form.
- For a clustered pattern, I is close to 1.
- For a regular pattern, I is close to -1.

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Geary's c

$$c = \frac{n-1}{2s_0} \times \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(x_i - x_j)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

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- Geary's c is a measure of dissimilarity.
- Under CSR, $E(c) = 1$ and the variance has a complicated form.
- Geary's C ranges from 0 to 2.

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- Under CSR, $E(c) = 1$ and the variance has a complicated form.
- Geary's C ranges from 0 to 2.
- For a **clustered pattern**, c is close to 0.
- For a **regular pattern**, $c > 1$.

Tests Based on Moran's I and Geary's c

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- **Approximate normal tests** or standard normal deviate under randomization.
- **Monte Carlo simulation** (or, permutation p-value).
- In general, Moran's I is more powerful than Geary's c, except in the case of binary data.

More on Neighborhood Structure

Recall the general form

$$G = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} y_{ij}$$

where $w_{ij} = 1$ if i, j are neighbors and $w_{ij} = 0$ if i, j are not neighbors or $i = j$.

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- Let $x_i = 1$ if a particular bird species is present on the ith island and $x_i = 0$ otherwise.

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- Account for **directions**.
- Note that w_{ij} may not be the same as w_{ji} in some cases.

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- Consider a 3×4 regular lattice with indexes

1	4	7	10
2	5	8	11
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- Consider a response variable $y =$ organic P (in ppm) in corn plants and an explanatory variable $x =$ inorganic P (in ppm) in soil in the ith plot for $i = 1, 2, \ldots, 12$.

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- The data are arranged in the tables below:

Inorganic P (ppm)
(explanatory variable)

0.7	1.2	3.6	19.0
1.0	0.8	18.5	27.8
1.4	0.9	21.9	20.9

Organic P (ppm)
(response variable)

0.1	3.9	5.1	36.3
1.8	2.0	26.2	38.6
2.3	1.3	30.1	28.4

Phosphorous Example A (cont'd)

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- Consider a linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where $\varepsilon_i \sim$ iid $N(0, \sigma^2)$, for $i = 1, 2, \ldots, n$.

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  where $\varepsilon_i \sim$ iid $N(0, \sigma^2)$, for $i = 1, 2, \ldots, n$.
- Recall the linear regression model assumptions.
  1. The model is correct.
  2. The errors are independent.
  3. The errors have a constant variance.
  4. The errors have normal distribution.

Phosphorous Example A (cont'd)

- Assume **a first-order neighborhood** and an **approximate normal test**.

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- For the organic P (y), Moran's I $= 0.59$ and p-value $= 0.0019$.

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- For the inorganic P (x), Moran's I $= 0.60$ and p-value $= 0.0017$.

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- For the organic P (y), Moran's I $= 0.59$ and p-value $= 0.0019$.
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- Suppose we went ahead and fitted a linear regression model.

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- For the inorganic P (x), Moran's I $= 0.60$ and p-value $= 0.0017$.
- Suppose we went ahead and fitted a linear regression model.
- For the residuals ($r_i = y_i - \hat{y}_i$), Moran's I $= 0.075$ and p-value $= 0.40$.

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- Q: Can we perform a linear regression on the data?

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- The spatial correlation in the inorganic P (x) and the close relation between organic P (y) and inorganic P (x) "causes" spatial correlation in the organic P (y).

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- Q: Can we perform a linear regression on the data?
- The spatial correlation in the inorganic P (x) and the close relation between organic P (y) and inorganic P (x) "causes" spatial correlation in the organic P (y).
- Consider also diagonal and second-order neighborhoods.

Phosphorous Example B

- Consider another possible data set:

Inorganic P (ppm)
(explanatory variable)

13.0	5.0	5.6	11.4
17.5	14.0	12.4	27.3
16.8	9.9	30.1	28.9

Organic P (ppm)
(response variable)

37.9	8.7	16.5	22.1
22.6	20.6	22.6	46.0
29.4	14.1	46.5	44.2

- Again, assume a first-order neighborhood and an approximate normal test.
- For the organic P (y), Moran's I $= 0.21$ and p-**value** $= 0.17$.
- For the inorganic P (x), Moran's I $= 0.15$ and p-**value** $= 0.27$.

Phosphorous Example B (cont'd)

- The residuals after a linear regression are:

-3.81	-1.39	2.66	4.24
-4.02	-0.99	3.30	5.29
-5.21	-1.60	1.77	-0.24

- Moran's I $= 0.60$ and p-value $= 0.0013$.

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- Moran's I $= 0.60$ and p-value $= 0.0013$.
- Q: Can we perform a linear regression on the data?
- Note: It is not enough to examine only the spatial patterns of x's or the y's.

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R Demo: Mosaic Example

Mosaic Example II

B	B	W	W
B	B	B	W
W	B	W	W

Numeric Coding

1	1	0	0
1	1	1	0
0	1	0	0

Coding scheme: $B = 1,\ W = 0$

Spatial Correlation in R

Ref: In-class demo and R Markdown file

- Define the spatial layout of the data
- Create neighborhood structure:
  - `cell2nb(..., type = "rook")`
  - `cell2nb(..., type = "queen")`
- Create spatial weights:
  - `nb2listw(..., style = "B")`
  - `nb2listw(..., style = "W")`
- Test for spatial correlation:
  - `geary.test()` for Geary's c
  - `moran.test()` for Moran's I
  - `moran.mc()` for Monte Carlo test

Main idea: define neighbors → define weights → run spatial test → interpret evidence against CSR

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Assignment 1 Solution

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Assignment 2

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Summary

- Spatial correlation and dependence
- Test for spatial correlation and model selection process based on spatial dependence.

Thank You